

OCC: SOFTWARE DEVELOPER YEAR 1

Automated Web Scrapping

DAG511

SAQA ID: 118707

Formative Assessment 1



CTU
training solutions

Private
Tertiary
Education

Table of Contents

Declaration of authenticity.....	3
QUALIFICATION AND ASSESSMENT DETAILS.....	4
INSTRUCTIONS.....	5
ASSESSMENT OUTCOME.....	5
QUESTION 1 (100).....	6
MARKING RUBRIC.....	9
Mark Allocation.....	10

Declaration of authenticity

I **Grant Samson** (FULL NAME)

hereby declare that the contents of this assessment **DAG511-FA1**

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any **AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.**

Activity	Date

Signature: 

Date: **02/05/2026**

QUALIFICATION AND ASSESSMENT DETAILS

Programme Title	OCC: Software Developer
Module Title	Automated Web Scraping
Module Code	DAG511
NQF Level	5
Assessment	Formative Assessment 1
Hand Out Date	29/04/2026
Hand in Date	07/05/2026
Time Allowed	2 Hrs
Total Marks	100
Developer	Kobus Mienie
Internal Moderator	Lerato Tsoehlisi
Internal Moderator	Chamaine Phoswa
External Moderator	

Student Name and Surname	Grant Samson
Student Number	20261424

INSTRUCTIONS

- Read each question carefully and consider the mark allocation before answering.
- Ensure you answer all questions.
- For final submission, include the Declaration of Authenticity, Pe-Assessment Document, Assessment document, and all the relevant files required to answer the questions in a zip folder.
- Only a .zip will be allowed for zipped folders
- Font style to be used Segoe UI 12pt, unless stated otherwise by a question
- Marks will be deducted for missing or incorrectly named documents

ASSESSMENT OUTCOME

- KM-03-KT01: Sources of data
- KM-03-KT02: Web scraping and access to accurate data
- KM-03-KT03: Big Data
- KM-03-KT04: Data quality
- KM-03-KT07: Wrangling

PRE-ASSESSMENT AGREEMENT

Assessment Preparation: Preparing the Candidate

Name of Candidate	Grant Samson	Date	08/06/2026
		Time	09:00 am
Name of Assessor	Mr NB Dube	Venue	STB
How to prepare the candidate	Document Requirements	Agree (tick)	Action Required
Explain to the candidate why you are meeting and the purpose of the assessment.	Assessment Policy Assessment process	☒	
Discuss the assessment plan in detail.	Assessment strategy	☒	
Explain assessment process, show assessment instruments to candidate and describe assessment conditions.	Assessment instruments	☒	
Identify the role-players during assessment.	Assessors Moderator	☒	
Describe the evidence required to be declared competent.	Examples of evidence	☒	
Explain how evidence will be judged.	Mark allocation explained	☒	
Explain to the candidate how to prepare: Give candidate assessment task description.	Assessment task description	☒	
Confirm with the candidate what he/she should bring to the assessment.	Detailed briefing on exact requirements to be given to candidate in writing	☒	
Ensure that candidate understands the procedures of all assessment practices.	Appeals Policy Appeals procedure Assessment Policy Assessment Procedure Moderation Policy Moderation procedure Verification Policy Verification Procedure	☒	
Ask the candidate if he/she foresees any problems or identify any special needs.	List needs	☒	

Agreed Assessment Plan

Candidate's Name:	Grant Samson		
Assessor's Name:	MR NB Dube		
Module name:	DAG511		
Assessment:	DAGF11 FA1		
Unit Standard/s:	• KM-03-KT01: Sources of data • KM-03-KT02: Web scraping and access to accurate data • KM-03-KT03: Big Data • KM-03-KT04: Data quality • KM-03-KT07: Wrangling		
Type of Assessment i.e. Formative assignment, Formative test, Formative Practical, Summative etc.	Formative Assessment		
Special Assessment Requirements:			
Event	Date, time and location	Resources required	Evidence to be generated
Assessments due dates.	14/05/2026	Assessments	Completed documentation
Complete activity on Campus Online/ upload to Campus Online		Study guide	Completed Portfolio of Evidence
Submit Electronic Portfolio of Evidence on Campus Online		Campus Online	Acknowledgement of receipt from Facilitator
Assessor Roles and Responsibility			
Roles	❖ Assessor ❖ Guide ❖ Feedback Agent ❖ Reviewer		
Responsibilities	❖ Consult candidate re-assessment, assessment process and plan. ❖ Agree assessment process and plan with candidate. ❖ Forward documentation to candidate: plan, guide and assessment instruments. ❖ Assess candidates with the use of different instruments.		

--

Candidate Roles and Responsibility		
Roles	❖ Leaner ❖ Feedback agent ❖ Reviewer	
Responsibilities	❖ Be available for assessment. ❖ Be actively involved in the consultative process. ❖ Learn from the assessment process. ❖ Provide feedback to the assessor in terms of the assessment as learning activity. ❖ Provide feedback to the assessor on the efficacy of the assessment process. ❖ Review own role and assessor role in the assessment process.	
Assessment Instruments	❖ Electronic Portfolio of Evidence ❖ Questionnaire ❖ Report ❖ Presentation ❖ Reflexive questions ❖ Work sample ❖ Practical's ❖ Group Activity ❖ Research activities	
Assessment Process		
	❖ Evaluation of PoE addressing Essential Embedded Knowledge in unit standards. ❖ Evaluation of Research Projects and other evidence addressing specific unit standards. ❖ Consultation: assessment plan and assessment activities and instruments. Pre-assessment moderation and interviews conducted at this stage. ❖ Observation: feedback on assessment against specific outcomes, critical outcomes in unit standards. ❖ Feedback: to candidate regarding sufficiency of evidence and possible interview to gain supplementary evidence. ❖ Feedback to candidate regarding assessment findings as well as review process.	☐ ☐ ☐ ☐ ☐ ☐
Feedback	Written feedback to be given to all stakeholders at the end of the assessment process, as well as verbal feedback to the candidate during assessment activities.	
Recording Process	Process and findings to be recorded and submitted for record keeping purposes as well as moderation and verification.	

Review Process	The review process is the responsibility of the assessor and the candidate.
----------------	---

	Joint reviewing will take place after feedback has been given to the candidate.
Right to appeal	The candidate must be advised of the right to appeal.
Resources Required	Assignments Campus Online Assessments OREILLY Guides

Overall Assessment Decision	Competent ☐	Not yet competent ☐	
Facilitator's Signature		Date	
Assessor's Signature		Date	
Moderator's Signature		Date	

QUESTION 1

(100)

Scenario

You are working as a junior data technician for a digital research company.

Your manager needs you to extract, clean, analyse, and store data from a public webpage, <https://quotes.toscrape.com>

You have been asked to build a Python script that:

- Fetches data from a given URL
- Extracts useful information
- Cleans and structures the data
- Saves it into CSV and JSON formats
- Generates a basic data summary

Part1: Project Setup & Data Fetching

(20)

1. Create a Python file called scraper.py (2)
2. Import the required libraries (5)
3. Write code to send a GET request to the chosen URL (6)
 - Handle possible errors using try/except
 - Check for status code 200
4. Save a copy of the raw HTML to a file (raw.html) (7)
 - File handling
 - Correctly write HTML content to file

Part 2: Data Extraction

(30)

5. Parse the HTML using BeautifulSoup (5)
6. Identify repeated data elements such as: (5)
 - Quotes

7. Extract at least 3 fields per item (12)

- Examples:
 - o Quote text
 - o Author
 - o Tags

8. Use regex at least once to clean or refine extracted data (8)

- Examples:
 - o Remove currency symbols
 - o Extract numbers
 - o Validate URLs
 - o Remove special characters

Part 3: Data Cleaning & Structuring (15)

9. Store extracted items in a list of dictionaries (8)

- Example:
 - o `{"quote": "...", "author": "...", "tags": "..."}`

10. Ensure all fields are clean and consistent (7)

- No blank fields
- No HTML tags
- Whitespace stripped

Part 4: Saving Output Files (20)

11. Export data to a CSV file (output.csv) (10)

- Proper header row
- Each dictionary written as a row

12. Export data to a JSON file (output.json) (10)

- Proper JSON structure
- UTF-8 encoding
- Indentation for readability

Part 5: Summary Report

(15)

13. Create a printed summary with at least 5 insights such as:

(15)

- Number of items scraped
- Most common author/tag/category
- Minimum/maximum price
- Average numeric value
- Count of unique values

End of Assessment

MARKING RUBRIC

Part	Description	Total Marks	Marks Awarded
1	Project Setup & Data Fetching	20	
2	Data Extraction	42	
3	Data Cleaning & Structuring	14	
4	Saving Output Files	19	
5	Summary Report	5	
TOTAL		100	

Mark Allocation

Task	Description	Marks
Question 1	Mini Web Scraper and Data Cleaning	100

LEARNING OUTCOMES INTEGRATED QUESTIONS CROSS-MAPPING GRID

LO No.	Learning Outcome Description	Question	Marks
KM-03-KT01	Sources of data	1	100
KM-03-KT02	Web scraping and access to accurate data		
KM-03-KT03	Big Data		
KM-03-KT04	Data quality		
KM-03-KT07	Wrangling		
TOTAL MARKS			100